

1. Administrative & Study Identification

Full Study Title [A Phase 2, Open-label, Study to Evaluate the Efficacy and Safety of Rapcabtagene Autoleucel in Patients With Active, Refractory Systemic Lupus Erythematosus (SLE) or Active, Refractory Lupus Nephritis (LN)] **Short Title/Acronym** [AUTOGRAPH - SLE/LN] **Protocol Number** [CYTB323J12201] **NCT Number** [NCT06581198] **Trial Status** [RECRUITING] **Sponsor/Company Name** [Novartis / Novartis Pharmaceuticals] **Investigational Product** [Rapcabtagene autoleucel (YTB323) - CD19-directed CAR-T cell therapy; Mapped Drug/Trade Name: ciltacabtagene autoleucel (Carvykti)] **ATC Code** [L01XL05] **Phase of Development** [PHASE2] **Medical Monitor** [Novartis Clinical Operations Lead]

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: [To evaluate the efficacy and safety of rapcabtagene autoleucel (administered once following lymphodepletion) in patients with active, refractory systemic lupus erythematosus (SLE) or active, refractory lupus nephritis (LN)]. **Secondary Objectives:** [To evaluate the percentage of participants achieving complete renal response (CRR), the number of weeks where Lupus Low Disease Activity Score (LLDAS) was achieved, and the percentage of participants without flaring (i.e., 1 new BILAG2004 A or 2 new BILAG2004 B flares)].

2.2 Background & Rationale [To address the critical unmet medical need for patients with active, difficult-to-treat (refractory) SLE and LN who have not responded to at least two prior standard therapies. The study utilizes rapcabtagene autoleucel, a novel chimeric antigen receptor (CAR) T-cell therapy designed to target and eliminate the B-cells that drive lupus disease activity and kidney inflammation, aiming for a deep, durable, and drug-free remission].

3. Study Design & Methodology

3.1 Design Framework Study Type: [INTERVENTIONAL] **Control Type:** [Single-arm / Standard of Care comparison (depending on specific study cohort)] **Blinding:** [Open-label] **Randomization:** [N/A / Randomized to SOC in some segments]

3.2 Planned Enrollment & Duration Target Enrollment: [179 participants] **Number of Sites:** [Global, including sites in the US, Australia, Austria, Brazil, Czechia, Denmark, France,

Germany, and Spain] **Estimated Study Start:** [04-Sep-2024]
Estimated Primary Completion: [14-Feb-2028]

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. [Men and women with SLE, aged ≥ 18 years and ≤ 75 years at screening, fulfilling the 2019 European League Against Rheumatism (EULAR)/American College of Rheumatology (ACR) classification criteria for SLE at screening]. 2. [Active lupus nephritis without signs of significant chronicity or active systemic lupus erythematosus. SLEDAI-2K score ≥ 6 points at screening, excluding points attributed to "fever", "lupus headache", "alopecia", and "organic brain syndrome"]. 3. [Participant must be positive for at least one of the following autoantibodies at screening: antinuclear antibodies (ANA) at a titer of $\geq 1:80$ (on HEp-2 cells or an equivalent positive test), anti-dsDNA (above the ULN), or anti-Sm (above the ULN) as determined by a central laboratory]. 4. [Inadequate response at screening to at least two therapies].

4.2 Exclusion Criteria 1. [Any acute, severe lupus-related flare at screening that needs immediate treatment other than pulse GCs and/or makes the immunosuppressive washout impossible and, thus, makes the participant ineligible for CD19 CAR-T therapy]. 2. [Inadequate organ function during screening and prior to randomization, or a history or current diagnosis of ECG or cardiac abnormalities indicating significant risk of safety for participants prior to randomization]. 3. [Human immunodeficiency virus (HIV) positivity at screening; acute or chronic infection with hepatitis B (HBV) or hepatitis C (HCV) at screening]. 4. [Grade 2 or higher thromboembolic event in the past 4 weeks prior to screening].

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: [Single infusion of rapcabtagene autoleucel (YTB323) administered once]. **Route of Administration:** [Intravenous (IV) infusion following pre-treatment lymphodepletion (using fludarabine and cyclophosphamide)]. **Duration of Treatment:** [One-time infusion, with monitoring up to 104 weeks and a long-term follow-up period of up to 15 years].

5.2 Concomitant & Prohibited Therapy Permitted: [Pulse corticosteroids as bridge therapy if strictly required; standard of care medications in SOC arms]. **Prohibited:** [Immunosuppressive therapies that interfere with T-cell collection or CAR-T cell expansion (washout periods required prior to leukapheresis and lymphodepletion)].

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
1	Up to 6 weeks	Informed Consent, Medical History, EULAR/ACR Criteria Check, Autoantibody Labs, Organ Function Tests
2	Days Prior to Infusion	Leukapheresis for T-cell collection, Lymphodepleting Chemotherapy (Fludarabine + Cyclophosphamide)
3	Day 0	Single IV Infusion of rapcabtagene autoleucel
4	Weeks 1 to 4	Intensive inpatient/outpatient safety monitoring for Cytokine Release Syndrome (CRS) and neurotoxicity
5	Weeks 24 to 52	Efficacy Assessment (DORIS criteria, CRR, LLDAS), Safety Labs

7. Outcome Measures (Endpoints)

7.1 Primary Outcomes Evaluate the efficacy of rapcabtagene autoleucel: Defined as: * [Meeting the criteria of the Definition Of Remission In Systemic Lupus Erythematosus (DORIS)] OR * [Achieving complete renal response (CRR)]

7.2 Secondary Outcomes 1. [Percentage of participants achieving complete renal response (CRR)] 2. [Number of weeks where Lupus Low Disease Activity Score (LLDAS) was achieved] 3. [Percentage of participants without flaring (i.e., 1 new BILAG2004 A or 2 new BILAG2004 B flares)]

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: [Continuous monitoring for CAR-T specific toxicities, including Cytokine Release Syndrome (CRS) and Immune Effector Cell-Associated Neurotoxicity Syndrome (ICANS)]. **Serious Adverse Events (SAEs):** [Reporting timeline according to standard Phase II regulatory requirements (typically 24 hours), particularly focusing on severe infections or cytopenias]. **Data Monitoring Committee (DMC):** [Independent Data Monitoring Committee established by Novartis to oversee trial safety].

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: [Full Analysis Set (FAS) for efficacy and Safety Set for patients receiving any fraction of the lymphodepleting regimen or CAR-T cells]. **Sample Size Justification:** [Targeting 179 participants to provide adequate statistical power to detect significant improvements in CRR, DORIS remission rates, and safety signals in a refractory population]. **Handling of Missing Data:** [Missing data handled via

standard imputation methods as pre-specified in the protocol for long-term efficacy tracking].

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki.

Monitoring Plan: [Intensive onsite monitoring during the lymphodepletion and immediate post-infusion period (first 4 weeks), followed by routine follow-up visits]. **Audit Trail:** [Electronic Data Capture (eCRF) with stringent audit trails for cellular product chain of identity and chain of custody].

11. Study Identifiers & Governance

Primary ID: [CYTB323J12201] **Posting ID:** [179478236634708606]

Registry Number: [NCT06581198] **Sponsor/Lead Organization:** [Novartis] **Study Title:** [A Study of Rapcabtagene Autoleucel in Active, Refractory Systemic Lupus Erythematosus (SLE) or Lupus Nephritis (LN) Patients (AUTOGRAPH - SLE/LN)] **Official Regulatory Title:** [A Phase 2, Open-label, Study to Evaluate the Efficacy and Safety of Rapcabtagene Autoleucel in Patients With Active, Refractory Systemic Lupus Erythematosus (SLE) or Active, Refractory Lupus Nephritis (LN)]

12. Clinical Framework (SLE/LN Specific)

Primary Condition / Disease Areas: [Lupus / Lupus Erythematosus, Systemic, Lupus Nephritis] **Preferred UMLS Name:** [Systemic Lupus Erythematosus] **MeSH Code:** [D008180] **SNOMED ID:** **HPO Code:** [HP: 0002725] **Diagnosis Threshold:** [2019 EULAR/ACR criteria; positive ANA ($\geq 1:80$), anti-dsDNA, or anti-Sm; SLEDAI-2K ≥ 6 excluding specific criteria points] **Medical Methodology:** [Autologous CD19-directed CAR-T cell therapy following cyclophosphamide/fludarabine lymphodepletion] **Optimization Target:** [Deep B-cell depletion leading to immune reset and long-term drug-free remission]

13. Study Procedures & Methodology

Management Pathway: [Leukapheresis $\rightarrow$ CAR-T Manufacturing $\rightarrow$ Lymphodepletion $\rightarrow$ Infusion $\rightarrow$ Monitoring] **Education Protocol (HCP):** [Site-specific training for CAR-T administration, CRS/ICANS grading, and toxicity management] **Education Protocol (Patient):** [Counseling on long-term follow-up requirements (up to 15 years) and infection risk] **Quality Audit Mechanism:** [Strict chain of custody for autologous cell manufacturing and tracking]

14. Observation & Follow-Up Schedule

Total Duration: [Up to 15 years (including standard 104-week study follow-up and subsequent long-term registry)] **Onsite Visit Cadence:** [Intensive during Weeks 1-4 post-infusion; then gradually transitioning to Q12W/Q24W] **Remote Monitoring Frequency:** [As needed based on patient status and local site protocols] **Checklist Review Interval:** [Regular safety and efficacy evaluation at defined protocol intervals (e.g., Week 24, Week 52)]

15. Sign-off & Approval

	Role		Name		Signature		Date			:---		:---		:---		:---	
	Principal Investigator		[Investigator Name]		_____		[DD-MMM-YYYY]			Clinical Operations Lead		[Operations Lead Name]		_____		[DD-MMM-YYYY]	
			Lead Statistician		[Statistician Name]		_____		[DD-MMM-YYYY]								